COURSE REFERENCE · VERSION 1.0

Radio Electronics: Components and Boards

Key formulas and rules of the course

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1. Basics: current, voltage, power

MODULE 1

Unit prefixes

k (kilo) = ×1000
M (mega) = ×1,000,000
m (milli) = ÷1000
μ (micro) = ÷1,000,000

EXAMPLE 4.7 kΩ = 4.7 × 1000 = **4700 Ω**; 250 mA = 250 ÷ 1000 = **0.25 A**

Ohm's law

I = U / R
U = I × R
R = U / I

I — current, A; U — voltage, V; R — resistance, Ω

Convert mA to A and kΩ to Ω before calculating.

EXAMPLE 12 V across 6 Ω: I = 12 / 6 = **2 A**

Power

P = U × I

P — power, W

EXAMPLE A drone on 4S (≈15 V) draws 20 A: P = 15 × 20 = **300 W**

2. Batteries and connectors

MODULE 1

LiPo battery voltage

$U = n \times U_{cell}$

n — number of cells (the number before S); 4.2 V full, 3.7 V nominal, ~3.5 V under load — time to land, below ~3.0 V — harmful

EXAMPLE 4S fully charged: 4 × 4.2 = **16.8 V**; nominal 4 × 3.7 = **14.8 V**

Capacity

Ah = mAh ÷ 1000

EXAMPLE 1300 mAh = **1.3 Ah**

Maximum battery current

I_{max} = capacity (Ah) × C

C is the number on the label (e.g. 75C). The real figure is often lower than promised.

EXAMPLE 1500 mAh, 75C: 1.5 × 75 = **112.5 A**

XT connectors

$I_{continuous}$ ≈ number in the name ÷ 2

The number in the name is the short-term (peak) current.

EXAMPLE XT60: 60 A peak, **30 A** continuous; XT30 — 15 A; XT90 — 45 A

3. Part markings

MODULES 2, 4

Resistor: colour bands

R = (digit 1, digit 2) × $10^{\text{multiplier}}$ Ω

black 0, brown 1, red 2, orange 3, yellow 4, green 5, blue 6, violet 7, grey 8, white 9. The tolerance band (gold ±5 %, silver ±10 %) stands apart — read from the other end.

EXAMPLE brown-black-orange: 1 and 0 → 10, plus three zeros = **10 kΩ**

SMD resistor: digit code

3 digits: two digits + number of zeros, Ω
4 digits: three digits + number of zeros, Ω
R is the decimal point: 4R7 = 4.7 Ω

EXAMPLE 472: 4 and 7 → 47, the third digit 2 means two zeros = 4700 Ω = **4.7 kΩ**

Capacitor: code (base unit — picofarads)

3 digits: two digits + number of zeros, pF
1 µF = 1000 nF = 1,000,000 pF

EXAMPLE 104: 1 and 0 → 10, plus four zeros = 100,000 pF = **100 nF**

Inductor: code (base unit — microhenries)

3 digits: two digits + number of zeros, µH
R is the decimal point in µH; N in nH

EXAMPLE 220 = 22 and no zeros = **22 µH**; 4R7 = **4.7 µH**; 2N2 = **2.2 nH**

SMD package size

code = length and width in hundredths of an inch; 1 inch = 25.4 mm

Size tells the power rating, not the value.

EXAMPLE 0603 = 0.06 × 0.03 in ≈ **1.6 × 0.8 mm**

78xx / 79xx regulators

78 — positive, 79 — negative; last two digits are the volts
L — up to ~0.1 A, M — up to ~0.5 A, no letter — up to 1.5 A

EXAMPLE 7912 = **−12 V**; 78L05 = **+5 V, up to 0.1 A**

Crystals and oscillators

frequency in MHz, the dot is a decimal point

Exception: the watch crystal 32.768 is in kHz.

EXAMPLE “40.000” = **40 MHz**; “12.000” = **12 MHz**

4. Basic circuits

MODULES 3, 6

Series and parallel

series: R = R1 + R2
parallel: the total is less than the smallest; two equal ones give half of one

Battery cells in series add voltages (that is how 6S is made).

EXAMPLE 1 kΩ + 2 kΩ = **3 kΩ**; two 10 kΩ in parallel = **5 kΩ**

Voltage divider

$U_{out} = U_{in} \times R2 / (R1 + R2)$

R1 — upper resistor (to supply), R2 — lower (to ground)

EXAMPLE 6S battery (25.2 V), R1 = 10 kΩ, R2 = 1 kΩ: 25.2 × 1 / (10 + 1) ≈ **2.3 V** — safe for the processor

LED resistor

$$R = (U_{supply} - U_{LED}) / I$$

Pick the nearest standard **higher** value.

EXAMPLE 5 V, red LED 2 V, 10 mA: R = (5 − 2) / 0.01 = 300 Ω → use **330 Ω**

RC circuit: time constant

$$\tau = R \times C$$

τ in seconds; R in Ω; C in farads. After τ the capacitor is ≈63 % charged, after 5τ almost fully.

EXAMPLE 10 kΩ × 1 µF = 10,000 × 0.000001 = **0.01 s**

Heating of an open MOSFET

$$P = I^2 \times R$$

R — on-resistance of the channel (hundredths to thousandths of an ohm)

EXAMPLE 20 A through 0.005 Ω: 20² × 0.005 = 400 × 0.005 = **2 W**

PWM: average voltage

$$U_{avg} = D \times U$$

D — duty cycle (the share of time the switch is on)

EXAMPLE 40 % of 16 V: 0.4 × 16 = **6.4 V**

5. Power supplies and motors

MODULES 4, 7, 8

Linear regulator: heat

$$P_{heat} = (U_{in} - U_{out}) \times I$$

A 78xx needs about 2 V more at the input than it puts out.

EXAMPLE 12 V → 5 V at 1 A: (12 − 5) × 1 = **7 W** of heat — needs a heatsink

Buck converter: duty cycle

$$D \approx U_{out} / U_{in}$$

EXAMPLE 16 V → 5 V: 5 / 16 ≈ **31 %**

Boost converter: output voltage

$$U_{out} \approx U_{in} / (1 - D)$$

EXAMPLE 3.7 V → 5 V: D ≈ 1 − 3.7 / 5 ≈ **26 %**; 5 V at D = 50 %: 5 / 0.5 = **10 V**

Power in and out

$$P_{in} \geq P_{out} \rightarrow I_{in} \geq U_{out} \times I_{out} / U_{in}$$

A converter does not create energy: the lower the input voltage, the higher the input current. Losses add more.

EXAMPLE A power bank giving 5 V / 2 A = 10 W from a 3.7 V cell: 10 / 3.7 ≈ 2.7 A — so **over 2.7 A**

Current sensor (shunt)

$$U = I \times R_{shunt} \rightarrow I = U / R_{shunt}$$

R0005 = 0.0005 Ω

EXAMPLE 100 A × 0.0005 Ω = **0.05 V** (50 mV)

Brushless motor speed

$$rpm \approx KV \times U$$

KV — rpm per volt with no load; marking 2207 = 22 mm stator, 7 mm height

EXAMPLE 1950 KV on 4S (16.8 V): 1950 × 16.8 ≈ **32,760 rpm** with no load

6. Radio

MODULES 11, 13, 14

Wavelength

$\lambda = 300 / f_{MHz}$ (metres)

A half-wave antenna ≈ λ / 2 (a little shorter in practice).

EXAMPLE 2400 MHz: 300 / 2400 = 0.125 m = **12.5 cm**; half ≈ **6.25 cm**

Tank circuit (LC)

$f = 1 / (2\pi \sqrt{(L \times C)})$

f in hertz; L in henries; C in farads

EXAMPLE L = 10 µH, C = 100 pF: 1 / (2π × √(0.00001 × 0.0000000001)) ≈ **5 MHz**

Power in dBm

$P_{mW} = 10^{dBm / 10}$

0 dBm = 1 mW; 20 dBm = 100 mW; 30 dBm = 1 W

EXAMPLE 27 dBm: $10^{2.7}$ ≈ **500 mW**

Decibel rules

+3 dB ≈ ×2
−3 dB ≈ ÷2
+10 dB = ×10

In decibels, multiplying becomes adding.

EXAMPLE 23 dBm = 20 + 3 → 100 mW × 2 = **200 mW**

Link budget

Received = Power + Antenna gains − Losses
Margin = Received − Sensitivity

all in dBm / dBi / dB; the link works while the margin is above zero

EXAMPLE 20 dBm + 2 dBi + 8 dBi − 120 dB = −90 dBm; sensitivity −105 dBm → margin **15 dB**

Distance and signal

×2 distance = ×¼ power = −6 dB

Works from any starting distance: 1 → 2 km just like 2 → 4 km. It is not "−6 dB per km". Free-space loss: L = 32.4 + 20·log f(MHz) + 20·log d(km) (ITU-R P.525).

EXAMPLE 2.4 GHz: 1 km ≈ 100 dB, 2 km ≈ 106 dB, 4 km ≈ 112 dB; a 15 dB margin after doubling the distance → **9 dB**

Jamming and signal (J/S)

$J/S = P_{jammer} / P_{signal}$ (at the receiver)
in dB: $J/S = P_{jammer}$ (dBm) − P_{signal} (dBm)

What matters is the ratio at the receiver, not the transmitters' output powers.

EXAMPLE Jammer −80 dBm, signal −90 dBm: J/S = −80 − (−90) = **+10 dB** — the jammer is ten times stronger

Sources: P. Horowitz, W. Hill. The Art of Electronics, 3rd ed., Cambridge University Press, 2015 · P. Scherz, S. Monk. Practical Electronics for Inventors, 4th ed., McGraw-Hill, 2016 · IEC 60062 — marking codes for resistors and capacitors · Manufacturer datasheets: STMicroelectronics L78 / L78L, Texas Instruments µA7800